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Dehumidification system

Abstract

A dehumidification system for removing water vapors from an air is disclosed. The system includes a dehumidification core defining an air channel and at least one vapor channel separated from the air channel by a membrane to facilitate a removal of moisture from the air flowing through the air channel and is selectively permeable to water and water vapor and impermeable to air. The dehumidification system further includes a water ejector having a throat portion fluidly coupled to the vapor channels and adapted to create a relatively lower pressure of water vapor within the vapor channels than in the air channel. The water ejector also includes an outlet portion disposed downstream of the throat portion and configured to increase the pressure of water to facilitate a condensation of the water vapors received from the vapor channels.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a 371 of international application no. PCT/US2021/070456 filed on Apr. 27, 2021, which claims priority to and the benefit of U.S. Provisional Patent Application Nos. 63/015,924, filed on Apr. 27, 2020; 62/704,864 filed on May 31, 2020, and 63/129,206 filed on Dec. 22, 2020, the contents of which are hereby incorporated by reference herein for all purposes.

TECHNICAL FIELD

(1) The disclosure relates to a dehumidification system having a dehumidification core to remove moisture from air.

BACKGROUND

(2) With changing climate, global warming, and urbanization there is an increase in demand of air-cooling systems for cooling homes and buildings. An air-cooling system, generally, include a refrigerant or a desiccant dehumidifier for removing heat and humidity from the air. Traditional air-conditioners requires high energy and refrigerant for removing heat can be harmful to the environment. Refrigerant based air conditioners and dehumidifiers manage sensible and latent loads in a combined step, as moisture is extracted through vapor condensation around heat exchangers during the refrigerant evaporation stage. This process is energy intensive and inefficient.

(3) Air coolers, such as, desert coolers, or cooling towers are also used for providing relatively cool air to the interior of an occupied space. These systems generally work on water evaporation principle and cool the air by evaporating water from a surface. However, these systems do not necessarily work well in the humid environment.

(4) U.S. Pat. No. 8,496,732 discloses an air-cooling system for dehumidifying air by establishing a humidity gradient across a water selective permeable membrane in a dehumidification unit. The humidity gradient is established by creating a vacuum on one side of the membrane using a vacuum pump creating a vacuum by compressing air. Due to the compressible nature of the air, creating vacuum by compressing air is not efficient and consumes more energy. Additionally, vacuum pumps creating vacuum by compressing air generally use a large fraction of non-condensable fluid, such as, dry air, for operation, and do not respond well in environments containing high condensable load, such as, water vapor or moisture. The air-cooling system includes a condenser to condense water vapors extracted from the air.

(5) Hence, there is still a need for improved air-cooling systems to facilitate removal of heat and humidity from the air, which is energy efficient and avoids the use of refrigerants.

SUMMARY

(6) In a first aspect, a dehumidification system for removing water vapors from an air is disclosed. The dehumidification system includes at least one dehumidification core defining an air channel and at least one vapor channel separated from the air channel. The dehumidification core further includes a membrane separating the at least one vapor channel from the air channel. The membrane is adapted to facilitate a removal of moisture from the air flowing through the air channel.

Moreover, the membrane is selectively permeable to water and water vapor and impermeable to air. Also, the membrane includes an ion exchange capacity of at least 1.0 milliequivalents per gram. The dehumidification system further includes a liquid ejector having throat portion and an outlet portion disposed downstream of the throat portion. The throat portion is fluidly coupled to the at least one vapor channel and is adapted to create a relatively lower pressure within the at least one vapor channel than in the air channel. The outlet portion is configured to increase the pressure inside the liquid ejector to facilitate a condensation of the water vapors received from the at least one vapor channel. The dehumidification system further includes a reservoir for storing a liquid and configured to receive the liquid from the liquid ejector, and a pump fluidly connected to the liquid ejector and the reservoir. The pump is configured to supply the liquid from the reservoir to the liquid ejector.

- (7) In a second aspect, the reservoir includes a drain valve for facilitating a drainage the liquid from the reservoir when a level of the liquid reaches above a first threshold level.
- (8) In some aspects, the reservoir is made of a thermally conductive material to facilitate a heat transfer from the liquid stored inside the reservoir to an ambient.
- (9) In some embodiments, the dehumidification system further includes a heat exchanger fluidly connected to the liquid ejector and configured to receive at least a portion of the liquid exiting the liquid ejector. The heat exchanger is configured to cool the received liquid.
- (10) In one embodiment, the heat exchanger is disposed upstream of the reservoir and supply the cooled liquid to the reservoir.
- (11) According to an embodiment, the heat exchanger is an air to liquid heat exchanger.
- (12) According to an embodiment, the membrane includes sulfonated block copolymers.
- (13) In one embodiment, the liquid ejector is a water ejector adapted to receive liquid water from the pump.
- (14) In one embodiment, the dehumidification system further includes a booster pump arranged between the dehumidification core and the liquid ejector to reduce the pressure inside the vapor channels to a pressure below a partial vapor pressure of water.
- (15) In one embodiment, the booster pump facilitates in reducing the pressure to a value less than equal to 31.7 mbarA at 25 degree Celsius.
- (16) In one embodiment, the booster pump facilitates in reducing the pressure to a value less than 150 mbarA.
- (17) In one embodiment, the booster pump facilitates in reducing the pressure to a value between 20 mbarA and 40 mbarA.
- (18) In one embodiment, the booster pump facilitates in reducing the pressure to a value between 10 mbarA and 20 mbarA.
- (19) In a second aspect, an air-cooling system is disclosed. The air-cooling system includes at least one dehumidification core defining an air channel and at least one vapor channel separated from the air channel. The dehumidification core further includes a membrane separating the at least one vapor channel from the air channel. The membrane is adapted to facilitate a removal of moisture from the air flowing through the air channel. Moreover, the membrane is selectively permeable to water and water vapor and impermeable to air. Also, the membrane includes an ion exchange capacity of at least 1.0 milliequivalents per gram. The air-cooling system further includes a water ejector having throat portion and an outlet portion disposed downstream of the throat portion. The throat portion is fluidly coupled to the at least one vapor channel, and is adapted to create a relatively lower pressure of water vapor within the at least one vapor channel than in the air channel. Further, the outlet portion is configured to increase the pressure of water to facilitate a condensation of the water vapors received from the at least one vapor channel. The air-cooling system further includes a reservoir for storing the water and configured to receive the water from the water ejector, and a pump fluidly connected to the water ejector and the reservoir. The pump is configured to supply the water from the reservoir to the water ejector. Furthermore, the air-cooling system has an evaporative cooler for cooling the air by facilitating an absorption of water into the air. The evaporative cooler is disposed downstream or upstream of the at least one dehumidification core.
- (20) In one embodiment, the air-cooling system further includes a reservoir adapted to receive the liquid water from the water ejector and stores the liquid water. The air-cooling system also includes a pump fluidly connected to the water ejector and the reservoir, and configured to supply the liquid water from the reservoir to the water ejector.
- (21) According to some embodiments, the reservoir includes a drain valve for facilitating a drainage the liquid water from the reservoir when a level of the liquid water reaches above a first threshold level.
- (22) In some embodiments, the reservoir is made of a thermally conductive material to facilitate a

heat transfer from the liquid water stored inside the reservoir to an ambient.

(23) In embodiments, the air-cooling system further includes a heat exchanger fluidly connected to the water ejector and configured to receive at least a portion of the liquid water exiting the water ejector. The heat exchanger is configured to cool the received liquid water.

(24) In some embodiments, the heat exchanger is disposed upstream of the reservoir and supply the cooled liquid water to the reservoir.

(25) According to one embodiment, the heat exchanger is an air to liquid heat exchanger.

(26) According to some embodiments, the membrane includes sulfonated block copolymers.

(27) In embodiments, the membrane includes an ion exchange capacity of at least 1.0 milliequivalents per gram.

(28) In one embodiment, the air-cooling system further includes a booster pump arranged between the dehumidification core and the water ejector to reduce the pressure inside the vapor channels to a pressure below a partial vapor pressure of water.

(29) In one embodiment, the booster pump facilitates in reducing the pressure to a value less than equal to 31.7 mbarA at 25 degree Celsius.

(30) In one embodiment, the booster pump facilitates in reducing the pressure to a value less than 150 mbarA.

(31) In one embodiment, the booster pump facilitates in reducing the pressure to a value between 20 mbarA and 40 mbarA.

(32) In one embodiment, the booster pump facilitates in reducing the pressure to a value between 10 mbarA and 20 mbarA.

(33) In some embodiments, the evaporative cooler is a first evaporative cooler disposed upstream of the dehumidification core and is adapted to cool the air flowing to the dehumidification core. The air-cooling system further includes a second evaporative cooler disposed downstream of the dehumidification core and configured to cool the air received from the dehumidification core.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of an embodiment of an air-cooling system having a dehumidification system.

(2) FIG. 2A illustrates a schematic view an embodiment of a dehumidification system having a dehumidification core having air channels separated from vapor channels by a membrane.

(3) FIG. 2B illustrates a schematic view an embodiment of a dehumidification system having a dehumidification core having air channels separated from vapor channels by a membrane.

(4) FIG. 3 is a schematic top view of a portion of an embodiment of a dehumidification core, having a flow of the water vapors from the air flowing through the air channel to the vapor channels.

(5) FIG. 4 is perspective view of an embodiment of a dehumidification core having air channels extending from a first face to a second face.

(6) FIG. 5 is sectional view of the dehumidification core of FIG. 4, depicting the vapor channels extending substantially perpendicular to a direction of extension of air channels.

(7) FIG. 6 is section view of an embodiment of a water ejector.

(8) FIG. 7 is an enlarged schematic view of an embodiment of a first evaporative cooler.

(9) FIG. 8 is an enlarged schematic view of an embodiment of a second evaporative cooler.

(10) FIG. 9 illustrates an assembled view of an embodiment of a stack of a dehumidification core.

(11) FIG. 10 illustrates an exploded view of the stack of FIG. 9.

(12) FIG. 11 illustrates a sectional view of an embodiment of a stack of a dehumidification core.

(13) FIG. 12 illustrates top view of the stack of FIG. 11 depicting an outer frame integrally formed

with a corrugated structure.

(14) FIG. 13 illustrates an embodiment of a stack of a dehumidification core.

(15) FIG. 14 illustrates a top sectional view of the stack of FIG. 13.

(16) FIG. 15 illustrates a top view of the stack of FIG. 13 depicting an outlet.

(17) FIG. 16 illustrates a perspective view of an embodiment of a dehumidification core.

(18) FIG. 17 illustrates a section view of the dehumidification core of FIG. 16 depicting a plurality of first corrugated structures defining a plurality of air channels, and a plurality of second corrugated structures defining a plurality of vapor channels separated from the air channels by a membrane.

(19) FIG. 18 illustrates a side view of FIG. 17 depicting vapor channels separated from the air channel via the membrane.

(20) FIG. 19 illustrates a side view of a dehumidification core formed using a conduit made of membrane.

(21) FIG. 20 illustrates a perspective view of the conduit of FIG. 19 in a coiled configuration.

DESCRIPTION

(22) The disclosure relates to dehumidification systems to remove moisture from air. In embodiments, the dehumidification systems facilitate removal of moisture from the air to reduce humidity level in the air supplied to a room with minimal consumption of energy. The dehumidification system includes at least one dehumidification core with at least a membrane adapted to facilitate a removal of moisture from the air flowing through an air channel. The dehumidification system includes a water ejector to create a vacuum inside a vapor channel to extract moisture from air flowing through the air channel. In embodiments, a booster pump is included to further reduce the pressure inside the vapor channels. The dehumidification system can be used for an air cooling system.

(23) Water (Liquid) Ejector: It is noted that the term “water ejector” is used in the disclosure here as water is a readily available media for use with the ejector, and water from vapor channel would separate in the tank and drained off that way. However, other liquids can be used with the ejector, e.g., oils of varying viscosities. A reference to “water” also includes other liquids as used in the disclosure herein.

(24) The water ejector works on a venturi effect and has a throat portion that has a relatively less diameter and through which the water is pumped. As the water flows through the throat portion, a velocity or a speed of the water increases, resulting into a low-pressure inside the throat portion. The low pressure is communicated to the vapor channels via a conduit. In this manner, the water ejector facilitates in creating a pressure differential between the air channels and the vapor channels of the dehumidification core to enable a flow of the moisture from the air flowing through the air channel to the vapor channels. To control a reduction of the pressure inside the throat portion, a velocity and a volume of the water entering an inlet portion of the water ejector is controlled.

(25) In embodiments, the water ejector may generate a vacuum pressure of 40 mbarA to 120 mbarA. It may be appreciated that vacuum generation capacity of the water ejector is limited by the saturation vapor pressure of the water. In embodiments, the water ejector may create a pressure of less than or equal to -960 mbar gage pressure.

(26) To further increase a vacuum level inside the vapor channels or decrease/reduce the pressure inside the vapor channels, the dehumidification system, in embodiments, includes a booster pump arranged between the dehumidification core and the water ejector.

(27) Booster Pump: The booster pump is a root pump that generally includes at least a pair of meshing lobes rotating in opposite directions. Fluid is trapped in pockets surrounding the lobes and carried from the inlet side to outlet side. The booster pump facilitates in decreasing the pressure inside the vapor channels to a pressure less than or equal to a partial vapor pressure of water.

(28) Partial vapor pressure of water at 25° C. is less 31.7 mbarA. In embodiments, the booster pump in combination with the water ejector facilitates a creation of pressure less than or equal to

20 mbarA inside the vapor channels. In some embodiments the booster pump facilitates in reducing the pressure to a value between 10 mbarA to 20 mbarA, or ≤ 10 mbarA, or between 20-40 mbarA, or ≤ 150 mbarA. The booster pump and the water ejector may be controlled to generate a vacuum (i.e., pressure) inside the vapor channels depending upon the amount of cooling and the water removal rate. For example, for a large system, such as a commercial air-cooling system and the dehumidification system wherein a relatively greater amount of water extraction rate is needed, the booster pump may facilitate in reducing pressure inside the vapor channels to less than 10 mbar. In such cases, multi-stage booster pump may be utilized. Alternatively, for a small establishment where a relatively less water vapor extraction rate is needed, a pressure corresponding to 10-20 mbar inside the vapor channel is sufficient. In such a case, a single stage booster pump may be utilized. Further, the speed of the booster pump may be controlled to vary the pressure inside vapor channels. By using the booster pump, the pressure differential between the air channels and the vapor channels is increased, resulting into increased water removal rate from the air flowing through the dehumidification core.

(29) Membrane: is characterized has having favorable ion-exchange capacity and proton conductivity, and glass transition temperature, providing both flexibility and material strength, and good stability and swelling properties even when hydrated. The membrane is formed mostly or substantially entirely from a sulfonated copolymer (SC) being sufficiently sulfonated to contain from 10-100 mol % sulfonic acid or sulfonate salt functional groups based on the number of monomer units in the copolymer. In embodiments, the SC is used to form a coating on a substrate surface with the substrate made of same or different material. In other embodiments, the membrane is used as a single or multiple SC layers or films, each with a certain or pre-selected thickness.

(30) In embodiments, the SC is a sulfonated block copolymer having a block copolymer molecular architecture with three or more blocks, designed to phase separate and form ion-conducting domains which enable water transmission, process that can be accelerated through application of voltage. In embodiments, the SC is selected from the group of perfluorosulfonic acid polymers such as sulfonated tetrafluoroethylene copolymer, polystyrene sulfonates, sulfonated block copolymers, polysulfones such as polyether sulfone, polyketones such as polyether ketone, and mixtures thereof.

(31) In embodiments, the sulfonated polymer is characterized as being sufficiently or selectively sulfonated to contain from 10-100 mol % sulfonic acid or sulfonate salt functional groups based on the number of sulfonatable monomer units in the sulfonated copolymer ("degree of sulfonation"). In embodiments, the sulfonated polymer has a degree of sulfonation of >25 mol %, or >50 mol %, or <95 mol %, or 25-70 mol %.

(32) In embodiments, the sulfonated polymer is characterized as having self-sterilizing effect, for killing at least 99% of microbes within 5 minutes of coming into contact with the coating material.

(33) In embodiments, the sulfonated polymer is a sulfonated block copolymer, having one or more copolymer block configurations corresponding to any of A-B-A, A-B-A-B-A, (A-B-A).sub.nX, (A-B).sub.nX, A-D-B-D-A, A-B-D-B-A, (A-D-B).sub.nX, (A-B-D).sub.nX or mixtures thereof, where n is an integer from 2 to about 30, X is a coupling agent residue, and wherein each D block is preferably a polymer block resistant to sulfonation. In embodiments, the SC has a linear structure corresponding to A-B-A, (A-B).sub.2X, A-B-D-B-A, (A-B-D).sub.2X, A-D-B-D-A, and (A-D-B).sub.2X, or a radial structure corresponding to (A-B).sub.nX and (A-D-B).sub.nX, where n ranges from 3 to 6. Two or more of the A, B, C, and D-blocks may be the same or different.

(34) In embodiments, the A-blocks are polymer segments of acrylic esters or methacrylic esters. In embodiments, the A-blocks are selected from polymerized para-substituted styrene monomers, ethylene, alpha olefins of 3 to 18 carbon atoms, 1,3-cyclodiene monomers, monomers of conjugated dienes having a vinyl content less than 35 mol percent prior to hydrogenation, acrylic esters, methacrylic esters, and mixtures thereof. If the A-blocks are polymers of 1,3-cyclodiene or conjugated dienes, the blocks are preferably hydrogenated after polymerization of the block

copolymer and before sulfonation of the block copolymer. If the A-blocks are hydrogenated polymers of 1,3-cyclodiene monomers, such monomers may be selected from the group consisting of 1,3-cyclohexadiene, 1,3-cycloheptadiene and 1,3-cyclooctadiene. The A-blocks may contain up to 15 mol percent of the vinyl aromatic monomers such as those present in the B blocks.

(35) The B blocks may contain from about 10 to about 100 mol percent sulfonic acid or sulfonate ester functional groups based on the number of monomer units, and comprises segments of one or more polymerized vinyl aromatic monomers selected from unsubstituted styrene monomer, ortho-substituted styrene monomers, meta-substituted styrene monomers, alpha-methylstyrene monomer, 1,1-diphenylethylene monomer, 1,2-diphenyl ethylene monomer, and mixtures thereof.

(36) The D-block may comprise a hydrogenated polymer or copolymer of a conjugated diene selected from isoprene, 1,3-butadiene, and mixtures thereof.

(37) X is a coupling agent residue, with the coupling agent selected from those known in the art, including polyalkenyl coupling agents, dihaloalkanes, silicon halides, siloxanes, multifunctional epoxides, silica compounds, esters of monohydric alcohols with carboxylic acids, (e.g. methylbenzoate and dimethyl adipate) and epoxidized oils.

(38) In embodiments, the SC is hydrogenated sulfonated block copolymer having the general configuration A-B, A-B-A, (A-B).sub.n, (A-B-A).sub.n, (A-B-A).sub.nX, (A-B).sub.nX or mixtures thereof, where n is an integer from 2 to about 30, and X is coupling agent residue. Prior to hydrogenation, each A block is a mono alkenyl arene polymer block and each B block is a controlled distribution copolymer block of at least one conjugated diene and at least one mono alkenyl arene. Subsequent to hydrogenation about 0-10% of the arene double bonds have been reduced, and at least about 90% of the conjugated diene double bonds have been reduced. Each A block has a number average molecular weight between about 3,000 and about 60,000. Each B block has a number average molecular weight between about 30,000 and about 300,000. Each B block comprises terminal regions adjacent to the A blocks that are rich in conjugated diene units and one or more regions not adjacent to the A blocks that are rich in mono alkenyl arene units. The total amount of mono alkenyl arene in the hydrogenated block copolymer is about 20 percent by weight to about 80 percent by weight. The weight percent of mono alkenyl arene in each B block is between about 10 percent and about 75 percent; at least 25% of the aromatic rings of the alkenyl arene are sulfonated. The hydrogenated sulfonated block copolymer has an ionic conductivity greater than 0.08 siemens/cm.

(39) Examples of SC that may be used are disclosed in published U.S. Pat. No. 8,222,346, and patent application nos. US20130108880A1 and US20140014289A1, incorporated herein by reference in their entirety. The SC may be prepared by anionic polymerization such as those disclosed in patent publications US20130108880A1 and US20140014289A1, incorporated herein by reference in their entirety. In various embodiments, the process may comprise polymerizing the suitable monomers in solution with a lithium initiator. The prepared block copolymers are sulfonated to obtain a sulfonated polymer product in solution and in micellar form. After sulfonation reaction, the block copolymers can be cast directly forming a membrane or film.

(40) In embodiments, the sulfonated copolymer is a sulfonated tetrafluoroethylene copolymer having a polytetrafluoroethylene (PTFE) backbone; side chains of vinyl ethers (e.g., —O—CF.sub.2—CF—O—CF.sub.2—CF.sub.2—) which terminate in sulfonic acid groups in a cluster region.

(41) In embodiments, the sulfonated polymer is a polystyrene sulfonate, examples include potassium polystyrene sulfonate, sodium polystyrene sulfonate, a co-polymer of sodium polystyrene sulfonate and potassium polystyrene sulfonate (e.g., a polystyrene sulfonate copolymer), having a molecular weight of >100,000 Daltons, >400,000 Daltons, and up to 1,500,000 Daltons. The polystyrene sulfonate polymers can either be crosslinked or uncrosslinked. In embodiments, the polystyrene sulfonate polymers are uncrosslinked and water soluble.

(42) In embodiments, the sulfonated polymer is a polysulfone, selected from the group of aromatic

polysulfones, polyphenylenesulfones, aromatic polyether sulfones, dichlorodiphenoxy sulfones, sulfonated substituted polysulfone polymers, and mixtures thereof. In embodiments, the sulfonated polymer is a sulfonated polyethersulfone copolymer, which can be made with reactants including sulfonate salts such as hydroquinone 2-potassium sulfonate (HPS) with other monomers, e.g., bisphenol A and 4-fluorophenyl sulfone. The degree of sulfonation in the polymer can be controlled with the amount of HPS unit in the polymer backbone.

(43) In embodiments, the sulfonated polymer is a polysulfone is a polyarylether ketone, e.g., a sulfonated polyether ketone (SPEEK), obtained by sulfonating a polyetherketoneketone (PEKK). The polyether ketone ketone is manufactured using diphenyl ether and a benzene dicarbonic acid derivative. The sulfonated PEEK is available as an alcohol and/or water-soluble product for subsequent used for coating membranes, for casting into membranes and films.

(44) With the use of the sulfonated copolymer, the SC membrane is hydrophilic and hygroscopic, and also permeable to moisture but not air and gases such as nitrogen and oxygen. The membrane containing sulfonated copolymer is characterized as selectively permeable with ion exchange properties. The SC membrane is also characterized as having excellent moisture vapor transport rates (MVTR) characteristics and excellent ionic exchange capacity.

(45) The SC membrane is also characterized as undergoing considerable swelling when it absorbs water, e.g., at least 100% at ambient temperature. In embodiments with the use of sulfonated block copolymer having a degree of sulfonation (e.g., at least 25 mol %), the SC membrane also shows anti-bacterial properties, making the AC particularly useful for sterilizing air in addition to cooling, in indoor spaces.

(46) In embodiments, the SC has an IEC of >0.5 meq/g, or 1.5-3.5 meq/g, or >1.25 meq/g, or >2.2 meq/g, or >2.5 meq/g, or >4.0 meq/g, or <4.0 meq/g.

(47) In embodiments, the SC membrane (film) or a coating containing SC has a thickness of >1 μm , or >5 μm , or 5-50 μm , or <100 μm , or <75 μm , or $<\mu\text{m}$. In embodiments, the membrane/coating may comprise nanocomposite materials and can have an average pore size <1 μm , or <0.5 μm , or <0.1 μm .

(48) In embodiments, the membrane is largely impermeable to air, e.g., having air permeability to be less than say less than 5 g/m²/day.

(49) In use, moisture from air can be extracted by a water permeable membrane by creating a pressure differential between the inlet side and the downstream end or section of a membrane assembly, which drives the diffusion of the water molecules from the inlet side towards the other side of the membrane.

(50) Depending on the unit, e.g., dehumidification unit or evaporative cooling unit, the SC membrane may be in forms other than sheets, e.g., netting, screen mesh or lattice, woven, non-woven, perforated or apertured plate, a foam, a hollow fiber membrane, or a pad having interconnecting interstices and passageways throughout the body, having the SC coated or bonded thereon. In embodiments, the SC membranes can be in spiral form or arranged in stacks, either parallel or perpendicular to the air stream direction.

(51) In embodiments, the SC membrane is in the form of hollow fibers. Humid air flows through the hollow fibers, which are under vacuum. The hollow fibers provide a large dehumidification surface area and may be oriented parallel or perpendicular to the air flow. When the inside of the hollow fiber membrane is placed under vacuum, an osmotic gradient is generated between each of the fiber's hollow core (which are substantially under vacuum) and the fiber's outer surface. In embodiments of hollow membranes, SC coatings or films may be applied on the inner, outer, or both inner and outer surfaces of the hollow fibers. Hollow fiber membranes are known in the art, e.g., as disclosed in U.S. Pat. No. 5,762,798, which is incorporated herein by reference.

(52) The membrane may comprise SC bonded or incorporated onto a frame, another one or more membranes, a polymer matrix, or a plurality of fiber bundles by processes known in the art, e.g., casting. The SC membrane can also be applied as a coating on a fiber matrix, or on the fan blades

in the evaporative cooler.

(53) The membrane can be bonded to a frame or another perforated layer serving as a support structure through which air and humidity can flow freely. A frame may comprise a metal or plastic and may be formed into any conceivable geometry, including, but not limited to, honeycomb and corrugated structures. In embodiments, the frame may have a honeycomb, spiral, non-woven or a plurality of porous designs for high surface area, with the proton conducting membrane being employed on multiple sides and with one side as opening for the air to flow in. In other embodiments, the frame is shaped as corrugated sheets with channels for increased exposed area. Depending on, for example, the amount of moisture to be removed or the size of the room, the number of membrane frames may also be varied by adding or taking out one or more frames.

(54) The form-retaining frame can be thermally or mechanically formed and are preferably rigid, semi-rigid, or substantially rigid. As used herein, a rigid, semi-rigid or substantially rigid frame is a frame comprising a material or structure able to maintain its shape under its own weight. Suitable frame materials include fiberglass, aluminum, carbon, or a rigid polymer based on polyester, polyethylene, polypropylene, polyethylene terephthalate, polyvinylchloride, a styrene/acrylonitrile/butadiene copolymer, nylon, polytetrafluoroethylene, aramid-based polymeric fibers, metal, metal alloys, cellulose, cellulose nitrate, cellulose acetate, and combinations thereof.

(55) A single frame may support a single membrane in each side, or the membrane assembly may comprise two or more membranes each supported by a frame connected in parallel in series to allow for a higher required voltage for mass fluxes on the order of $10 \text{ sup.}^{-9} \text{ kg/s-cm.sup.2}$. In one embodiment, a 5-ton air conditioning unit with a sensible heat ratio of about 0.6 would require less than 30 m.sup.2 of SC membrane with IEC of at least 0.5, or >0.75 , or $>1.0 \text{ meq/g}$.

(56) The frame preferably has a thickness sufficient to maintain strength and without interfering with air flow or moisture transport. The frame thickness may range from about 25 microns to about 500 microns, from about 100 microns to about 500 microns, or from 200 microns to 500 microns, or at least 300 microns. The thickness typically depends on multiple factors including the number of stacked layers of frames used, air flow rate, and pressure.

(57) A frame may be porous with pore size sufficient to permit direct air contact of without interfering with moisture transport or significant pressure drop. The pore diameter can range from 0.1 to 200 microns, such as around 5 microns, up to around 8-10 mm, or larger, with the frame surface between the pores sufficient for a film or coating comprising, for example, selectively permeable and ion exchanging polymer, to be bonded onto the frame and maintain film attachment.

(58) When SC polymer is to be applied onto membranes, matrix, fibers, etc., as a coating, the coating can be prepared using different base solvents depending on, for example, the desired coating thickness or the membrane's intended application, e.g., whether the membrane is for a dehumidifier or evaporative cooler.

(59) In embodiment of an evaporative cooler, a SC-mediated evaporative cooling process is achieved through the evaporation of water supplied to the evaporative cooler membrane, which serves as a substrate that allows thermal energy transfer between the relatively warm, dehumidified flowing air and the cooler water molecules at or near the membrane surface over which the warm, dehumidified air flows.

(60) The thermal energy from the relatively warm flowing air molecules near the membrane surface can be absorbed by the cooler water molecules at or near the membrane's surface, which leads to cooling of the flowing air and evaporation of the water molecules from the membrane surface. Thermal energy transfer may also occur through collisional energy transfer between the relatively cooler (lower energy) water vapor molecules that evaporated from the membrane surface and the relative warm (higher energy), flowing air molecules. Also, a vapor pressure differential on opposite membrane surfaces facilitate water evaporation from the lower vapor pressure side of the membrane.

(61) Reference will be made to the figures, showing various embodiments of the fixture and

methods for using thereof.

(62) Referring to FIG. 1, an air-cooling system **100** having a dehumidification system **102** according to an embodiment is shown. The air-cooling system **100** receives an incoming air stream (hereinafter referred to as a first air stream) **200A** into a room **300** and delivers or generates an air having relatively less moisture content and a reduced temperature (hereinafter referred to as a second airstream **200B**). The air-cooling system **100** includes the dehumidification system **102** for facilitating a removal of moisture (i.e., water vapors) from the air delivered into the room **300**, and at least one evaporative cooler, for example, a first evaporative cooler **104** and a second evaporative cooler **106**, for cooling (i.e., reducing a temperature) the air delivered into the room **300**.

(63) As shown in FIG. 1, FIG. 2A and FIG. 2B, the dehumidification system **102** includes a dehumidification core **110** that receives an air **400A** having relatively high moisture content (also referred to as inlet air **400A**) and discharges an air **400B** having relatively low moisture content (also referred to as outlet air **400B**). For so doing, referring to FIGS. 2A, 2B, 3, 4, and 5, the dehumidification core **110** defines at least one air channel **114** through which the inlet air **400A** flows, and at least one vapor channel **116** that receives moisture (e.g., water vapors **120** extracted from the inlet air **400A**) flowing through the at least one air channel **114**. As shown, each vapor channel **116** is disposed adjacent to one or more of the air channels **114** and is separated by a membrane **122**. Conversely, each air channel **114** is disposed adjacent to one or more of the vapor channels **116** and is separated from the adjacent vapor channels **116** by the membrane **122**.

(64) In embodiments, as shown in FIG. 4, the dehumidification core **110** has a box structure, with a first face **126** disposed substantially perpendicular to the flow of inlet air **400A** and defining an inlet **128** for each of the air channel **114**, and a second face **130** disposed substantially parallel to the first face **126** and defining an outlet **132** of each of the air channel **114**. Accordingly, the inlet air **400A** enters into the air channels **114** through the first face **126** and exits the dehumidification core **110** through the second face **130** as the outlet air **400B**. The moisture (i.e., the water vapors **120**) so removed from the inlet air **400A** is collected into the vapor channels **116** (as shown in FIG. 3).

(65) As shown, each vapor channel **116** may extend in a direction substantially perpendicular to a direction of extension of the air channels **114**, and may extend in a direction substantially perpendicular to a third face **134** (i.e., top face **134**) of the dehumidification core **110** and a fourth face **136** (i.e., bottom face **136**) of the dehumidification core **110**. Further, the fourth face **136** may define a closed end of each of the vapor channel **114**, thereby preventing an exit of the water vapors **120** from the dehumidification core **110** through the bottom face **136**, while an outlet **138** (shown in FIG. 5) for each vapor channel **114** may be disposed proximate to the top face **134** to facilitate an exit of the water vapors **120** from the vapor channels **116**. In embodiments, the dehumidification core **110** may include one or more collection channels (not shown) disposed proximate to the top face **134** and in fluid communication with the outlets **138** of the vapor channels **116**. In such a case, the outlets **138** may be defined by an intermittent surface of the dehumidification core **110** that may be disposed proximate to the top face **134** and extends parallel to the bottom face **136**. In some embodiments, the one or more collection channels (not shown) may extend substantially parallel to the air channels **114** and may include closed ends at the first face **126** and the second face **130**.

(66) Also, as shown in FIG. 4 and FIG. 5, the dehumidification core **110** is formed by arranging a plurality of stacks **141** such the stacks **141** are arranged parallel to and spaced apart from each other to define air channels **114** therebetween. The stacks **141** are arranged in a manner such that membrane **122** of one stack **141** faces the membrane **122** of the adjacent stack **141**. Further, each stack **141** includes an outer frame **142** and a corrugated structure **143** disposed inside the outer frame **142** and supported by the outer frame **141**. As shown, the outer frame **142** is attached to an outer edge of the corrugate structure **143**. Further each corrugated structure **143** defines the plurality of vapor channels **116** having a substantially rectangular shape. Further, each stack **143**

includes two membranes **122**, one membrane **122** disposed at a first side of the corrugated structure **143**, and other membrane **122** disposed on second side, opposite to the first side, of the corrugated structure **143**. Accordingly, the corrugated structure **143** is arranged between the two membranes **122**, and each membrane **122** abuts the corrugated structure **143**. Accordingly, the membrane **122** separates the vapor channels **116** of the corrugated structure **143** from the air channel **114** defined between two adjacent stacks **141**.

(67) Although a box, e.g., cuboidal structure having the vapor channels **116** extending substantially perpendicular to the air channels **114** is contemplated, it may be appreciated that the dehumidification core **110** may include any other suitable shape or structure known in the art. Further, it may be envisioned that the air channels **114** and the vapor channels **116** may extend substantially in a same direction and may run parallel to each other. Also, the dehumidification core **110** having concentric air channels **114** and vapor channels **116** may also be envisioned. Further, the dehumidification core **110** may include a conduit **140** (shown in FIGS. **1**, **2**, and **4**) in fluid communication with each vapor channel **116** to receive the water vapors **120** and facilitates an exit of water vapors **120** from the dehumidification core **110**.

(68) As shown in FIG. **3**, the movement of the water vapors **120** from the air flowing through the air channels **114** to the vapor channels **116** is facilitated by the membrane **122**, while flow of other components **144**, for example, nitrogen, oxygen, carbon dioxide, etc., of the air from the air channels **114** to the vapor channels **116** is substantially blocked by the membrane **122**. In some embodiments, the membrane **122** may block a flow of about 99 percent of the other components **144** from the air channels **114** to the vapor channels **116**. In certain implementations, the membrane **122** may block a flow of about 95 percent to 99 percent of other components **144** from the air channels **114** to the vapor channels **116**.

(69) The membrane **122** facilitates an extraction of the water vapors **120** from the air flowing through the air channels **114** and flow of the water vapors **120** into the adjacent vapor channels **116**, through the membrane **122**, in response to a presence of relatively low pressure inside the vapor channels **116** than the air channels **114**. As such, a humidity gradient is established between the air channels **114** and adjacent vapor channels **116**. The humidity gradient is generated by creating a pressure gradient/differential between the air channels **114** and adjacent vapor channels **116**. In particular, a partial pressure of water vapor inside the vapor channels **116** is maintained at a level lower than a partial pressure of water vapor inside the air channels **114** to pull the water vapors **120** in the air flowing through the air channels **114** toward the suction side (i.e., the vapor channels **116**).

(70) Again referring to FIG. **1** and FIG. **2A**, to create/maintain a relatively lower pressure inside the vapor channels **116**, the dehumidification system **102** may include a pressure reducer system **146** having a liquid ejector, such as, a water ejector **150**, fluidly coupled to the vapor channels **116** via the conduit **140**, a pump **152** to supply liquid water to the water ejector **150**, and a reservoir **154** to store the liquid water and receive the liquid water expelled from the water ejector **150**. As shown, the water ejector **150** is in fluid communication with the vapor channels **116** via the conduit **140** extending from the dehumidification core **110** to the water ejector **150**. Accordingly, the water vapors **116** extracted from the inlet air **400A** flow towards the water ejector **150** via the conduit **140** in response to creation of low pressure in the water ejector **150**. As best shown in FIG. **6**, the water ejector **150** includes an inlet portion **158** defining an inlet port **160** of the water ejector **150**, a throat portion **162** extending from the inlet portion **158** in a longitudinal direction, and an outlet portion **164** extending from the throat portion **162** and defining an outlet port **166** of the water ejector **150**. Also, the water ejector **150** defines a vapor inlet port **168** coupled to the conduit **149** and in fluid communication with the throat portion **162** to facilitate a creation/maintenance of relatively low pressure (i.e., vacuum) inside the vapor channels **116**, and an entry of the water vapors **120** inside the water ejector **150** (i.e., the throat portion **162**) from the vapor channels **116**.

(71) As shown in FIG. **6**, the inlet portion **158** may include a nozzle portion **170** defining an orifice

172 to inject/supply the liquid water to the throat portion **162** at a relatively high speed, while a cross-sectional area of the outlet portion **164** increases, gradually, from the throat portion **162** to the outlet port **166** to decrease the speed to the liquid water received from the throat portion **162**. In some embodiments, a cross-sectional area of the inlet portion **158** may decrease, gradually, from the inlet port **160** to the throat portion **162** to facilitate a gradual increase in velocity/speed of the liquid water. Accordingly, the inlet portion **158** facilitates in increasing the velocity/speed of the liquid water as the liquid water flows from the inlet port **160** to the throat portion **162**, while the outlet portion **164** is configured to decrease the velocity/speed of the liquid water as the liquid water flows from the throat portion **162** to the outlet port **166**. Consequently, the pressure of the liquid water at the throat portion **162** is lower relative to the pressure of the water at the inlet port **160** and the outlet port **166**. A level of the pressure (i.e., vacuum) at the throat portion **162** may be adjusted/controlled by controlling a velocity or amount of water entering the inlet portion **158**. As such, the throat portion **162** is in fluid communication with the vapor channels **116** of the dehumidification core **110**, therefore, a vacuum (i.e., reduced pressure) is created inside the vapor channels **116**. The level of pressure at the throat portion **162** is controlled in such a manner that the reduced pressure created inside the vapor channels **116** is lower than the pressure in the air channels **114** by a desired value. Accordingly, the velocity of the liquid water entering the water ejector **150** is controlled so as to maintain the pressure inside the vapor channels **116** lower than the pressure of air inside the air channels **114** by the desired value so as to facilitate an extraction and flow of the water vapors **120** from the inlet air **400A** to the vapor channels **116** through the membrane **122**. However, the reduced pressure created/maintained inside the vapor channels **116** may be higher than the pressure inside the throat portion **162** due to loss of vacuum along the conduit **140**.

(72) In order to control and provide a flow of water to the water ejector **150**, the dehumidification system **102** includes the pump **152** disposed upstream of the water ejector **150** and fluidly connected to the inlet port **160** via first pipe **174**. In embodiments, the pump **152** may be a variable displacement pump to allow a control of the amount of water being pumped. Further, the pump **152** may be connected to the reservoir **154**, via the second pipe **176**, to receive the liquid water from the reservoir **154** and provide the liquid water at a desired velocity/speed to the water ejector **150**. In embodiments, the reservoir **154** may include a plurality of fins (not shown) along an outer surface of the reservoir **154** to facilitate a transfer of heat between the liquid water stored inside the reservoir **154** and ambient. In some embodiments, the reservoir **154** is made of a material having high thermal conductivity to facilitate a heat transfer between the liquid water stored inside the reservoir **154** and the ambient. Also, the reservoir **154** is fluidly connected to the outlet portion **164** (i.e., the outlet port **166**) of the water ejector **150** and is adapted to receive the liquid water exiting the water ejector **150**. Further, the reservoir **150** may include a drain valve **180** adapted to facilitate a draining of liquid water from the reservoir **154**. The drain valve **180** is adapted to move to an open position to drain the water out of the reservoir **154** when a level of the liquid water reaches beyond a first threshold level. In this manner, the drain valve **180** facilitates in maintaining the level of liquid water inside the reservoir **154** below the first threshold level.

(73) In some embodiments, at least a portion of the liquid water exiting the water ejector **150** is cooled down before being supplier to the reservoir **154**. For so doing, the pressure reduce system **146**, and hence the dehumidifier system **110**, may include a heat exchanger **182** to cool (i.e., reduce a temperature) the liquid water expelled by the water ejector **150**. The heat exchanger **182** may be an air to liquid heat exchanger, and may be disposed downstream of the water ejector **150** and upstream of the reservoir **154** and adapted to receive the liquid water from water ejector **150** and supply cooled liquid water to the reservoir **154**. As shown, the heat exchanger **182** is fluidly connected to the outlet port **166** vis a third pipe **184** and receive the liquid water from the water ejector vis the third pipe **184**. Similarly, the heat exchanger **182** is fluidly connected to the reservoir **154** via a fourth pipe **186** and supply the cooled liquid water to the reservoir **154** via the fourth pipe

186. In some implementations, only a portion of the liquid water expelled from the water ejector **150** is provided to the heat exchanger **182** by a bypass conduit. In embodiments, the portion of the liquid water, cooled by the heat exchanger **182**, is mixed with a remaining portion of liquid water before the liquid water enters the reservoir **154**. In some implementations, the portion of liquid water, cooled by the heat exchanger **182**, may directly flow/enters the reservoir **154**. In embodiments, the pressure reducer system **146** may include a blower **188** to increase a flow of air towards the heat exchanger **182** to facilitate the cooling of the liquid water flowing through the heat exchanger **182**. The blower **188** may be disposed upstream or downstream, in a direction of flow of air) of the heat exchanger **182**. In embodiments, a heat exchanger may be disposed between the water ejector **150** and the dehumidification core **110** to facilitate a cooling of the water vapors exiting the dehumidification core **100**. In such a case, the heat exchanger **182** and the associated blower **188** may be omitted.

(74) Referring to FIG. 2B, a dehumidification system **102'** having a pressure reducer system **146'** according to an alternative embodiment is shown. The pressure reducer system **146'** is similar to the pressure reducer system **146** except that a booster pump **148** is arranged between a water ejector **150'** and the dehumidifier core **110**. The booster pump **148** enables a further reduction of pressure, hence increase in vacuum, inside the vapor channels **116** of the dehumidification core **110**. In embodiments, the booster pump **148** may facilitate in reducing and maintaining the pressure inside the vapor channels **116** below a threshold value. In some embodiments, the threshold value corresponds to a saturation vapor pressure of water. Accordingly, the booster pump **148** facilitates in increasing a pressure differential between the vapor channels **116** and the air channels **114** to increase the water extraction rate from the air flowing through the air channels **114**. In embodiments, the booster pump **148** and the water ejector **150'** together reduce the pressure at the inlet of the booster pump **110** to a value between 20 mbarA to 40 mbarA. In embodiments, the pressure reducer system **146'** may create a pressure of less than 20 mbarA at the inlet of the booster pump **148**. In embodiments, the pressure reducer system **146'** may create a pressure of less than 10 mbarA at the inlet of the booster pump **148**. In embodiments, the pressure reducer system **146'** may create a pressure between 10 mbarA and 20 mbarA at the inlet of the booster pump **148**. The booster pump **148** and the water ejector **150'** may be controlled to reduce the pressure inside the vapor channels **116** depending upon a size of the dehumidification system **102'** and a desired water extraction rate.

(75) In embodiments, the booster pump **148** is a root pump. However, it may be appreciated that any type of vacuum booster pump known in the art may be utilized. Additionally, the water ejector **150'** is similar to the water ejector **150** except that a vapor inlet port **168'** that is in fluid communication with the booster pump **148** is arranged proximate to an inlet port **160** of an inlet portion **158** of the water ejector **150'** rather than a throat portion **162** of the water ejector **150'**.

(76) Again referring to FIG. 1, the first evaporative cooler **104** is disposed upstream of the dehumidifier core **110** and is arranged to receive the first air stream **200A** from the room **300**, while the second evaporative cooler **106** is disposed downstream of the dehumidifier core **110**. As shown in FIG. 1 and FIG. 7, the first evaporative cooler **104** includes an evaporative pad **500** (herein after referred to as a first evaporative pad **500**), a fan **502** (hereinafter referred to as first fan **502**) disposed upstream of the first evaporative pad **502** to draw the first airstream **200A** from the room **300** towards the first evaporative pad **502**. The first evaporative cooler **104** may further include a pan **504** (hereinafter referred to as a first pan **504**) to collect any excess water drained from the first evaporative pad **500**. The first evaporative pad **500** facilitates an evaporation of liquid water into the first airstream **200A** passing through the first evaporative pad **200A**, and thereby facilitates in cooling of the first airstream **200A**. As such, the liquid water absorbs the latent heat from the first airstream **200A** and gets converted into water vapors as the air (i.e., the first air stream **200A**) passes through the first evaporative pad **500**. Accordingly, the air gets cooled as the air passes through the first evaporative pad **500**, while a moisture content of the air increases. In this manner,

the first evaporative cooler **104** receives the first airstream **200A** from the room **300** and expels the air having relatively high humidity and lower temperature as the inlet air **400A**.

(77) In embodiments, the first evaporative pad **500** may include a honeycomb structure **506** (shown in FIG. 7) to provide a relatively large surface area to facilitate an evaporation of the liquid water into the air. Although the honeycomb structure **506** is contemplated, it may be appreciated that the first evaporative pad **500** may include any other structure suitable for facilitating the evaporation of the liquid water. Further, the first evaporative cooler **104** may include one or more nozzles (not shown) to inject liquid water into the first evaporative pad **500** and a pump **508** (shown in FIG. 7) to provide the liquid water from the first pan **504** to the one or more nozzles (not shown) to keep the first evaporative pad **500** wet. Moreover, the first fan **502** helps in pulling the first airstream **200A** from the room **300** towards the first evaporative pad **500** may also be disposed downstream of the first evaporative pad **500**. In such a case, the first fan **502** may be disposed upstream of the dehumidifier core **110**. Although the air-cooling system **100** having a single evaporative cooler **500** disposed upstream of the dehumidifier core **110** is shown and contemplated, it may be envisioned that the air-cooling system **100** may include any number of evaporative coolers disposed in series and upstream of the dehumidifier core **110**.

(78) As shown in FIG. 1, the second evaporative cooler **106** is arranged to receive an air (i.e., the outlet air **400B**) discharged by the dehumidification core **110** and provides a relatively cooler air (i.e., the second airstream **200B**) to the room **300**. As shown in FIG. 1 and FIG. 8, the second evaporative cooler **106** may include an evaporative pad **600** (hereinafter referred to as second evaporative pad **600**), a fan **602** (hereinafter referred to as second fan **602**) disposed upstream of the second evaporative pad **600** to direct and supply the cooled air (i.e. the second airstream **200B**) to the room **300**, and a pan **604** (herein after referred to as second pan **604**) to collect excess water drained from the second evaporative pad **600**. The second evaporative pad **600** facilitates an evaporation of liquid water into the air (i.e., the outlet air **400B**), and thereby facilitates in cooling of the air passing through the second evaporative pad **600**. As such, the liquid water absorbs the latent heat from the air and gets converted into water vapors as the air passes through the second evaporative pad **600**. Accordingly, the air gets cooled as the air passes through the second evaporative pad **600**, while a moisture content of the air also increases. In this manner, the second evaporative cooler **106** receives the outlet air **400B** discharged by the dehumidifier core **110** and expels and direct the air having relatively high humidity and lower temperature as the second airstream **200B** to the room **300**.

(79) In embodiments, the second evaporative pad **600** may include a honeycomb structure **606** (shown in FIG. 8) to provide a relatively large surface area to facilitate an evaporation of the liquid water into the air. Although the honeycomb structure **606** is contemplated, it may be appreciated that the second evaporative pad **600** may include any other structure suitable for facilitating the evaporation of the liquid water. Further, the second evaporative cooler **106** may include one or more nozzles (not shown) to provide the liquid water into the second evaporative pad **500** and a pump **608** (shown in FIG. 8) to provide the liquid water from the second pan **604** to the one or more nozzles (not shown) to keep the second evaporative pad **500** wet. Although the air-cooling system **100** having a single evaporative cooler **600** disposed downstream of the dehumidifier core **110** is shown and contemplated, it may be envisioned that the air-cooling system **100** may include any number of evaporative coolers disposed in series and downstream of the dehumidifier core **110**. Although two evaporative coolers **104**, **106** are contemplated, it may be appreciated that any one of the first evaporative cooler **104** and the second evaporative cooler **106** may be omitted.

(80) Furthermore, in some embodiments, the air-cooling system **100** may include a controller and a plurality of sensors for controlling an operation of the air-cooling system **100**. In embodiments, the air-cooling system **100** may include one or more temperature sensors and one or more humidity sensors to monitor a temperature and a humidity of at least one of the first airstream **200A**, the inlet air **400A**, the outlet air **400B**, and the second air stream **200B**. Accordingly, the controller may

control the pump **152** to deliver the liquid water at an optimum speed so as to maintain or create desired level of vacuum or pressure inside the vapor channels **116**. Similarly, the controller may control an amount of water injection into the first evaporative pad **500** and/or the second evaporative pad **600** based on one or more inputs received from the sensors.

(81) In embodiments, as shown in FIG. **1**, the air-cooling system **100** may be a split type air-cooling system having first unit **700** installed inside the room **300** and a second unit **702**, as an outdoor unit, disposed outside the room **300**. As shown, the first unit **700** includes a housing **704** (hereinafter referred to as first housing **704**) defining an inlet opening **706** for facilitating an entry of relatively hot and humid air (i.e., the first airstream **200A**) inside the first housing **704** from the room **300**, and an outlet opening **708** to facilitate an exit of relatively cold and dry air (i.e., the second airstream **200B**) to the room **300** from the first housing **704**. As shown in FIG. **1**, the first unit **700** may include the first evaporative cooler **104**, the dehumidifier core **110**, and the second evaporative cooler **106** mounted inside the first housing **704**. The first evaporative cooler **104** may be mounted inside the first housing **704** and disposed proximate to the inlet opening **706**, while the second evaporative cooler **106** may be mounted proximate to the outlet opening **708**. Moreover, the dehumidification core **110** is mounted to the first housing **704** and is between the first evaporative cooler **104** and the second evaporative cooler **106**.

(82) Further, the second unit **702** may include a second housing **710** and the water ejector **150**, the pump **152**, the reservoir **154**, and the heat exchanger **182**. The water ejector **150**, the pump **152**, the reservoir **154**, and the heat exchanger **182** are disposed inside the second housing **710** and mounted to the second housing **710**. Further, the conduit **140** extends outside the first housing **700** and connected to the vapor inlet port **168** of the water ejector **150**. Although the air-cooling system **100** is contemplated as the split air-cooling system, it may be appreciated that air-cooling system **100** may be contemplated as single window type air-cooling system. In such a case, the second housing **710** may be omitted, and the water ejector **150**, the pump **152**, the reservoir **154**, and the heat exchanger **182** may be mounted inside the first housing **700**.

(83) Although, the air-cooling system **100** having a single dehumidification system **102** is shown and contemplated, it may be appreciated that the air-cooling system **100** may include any number of dehumidification system **102** arranged in series configuration, or a parallel configuration, or a combination thereof. Further, the dehumidification system **102** having a single dehumidification core **110** is shown and contemplated, it is possible that the dehumidification system may include multiple dehumidification cores **110** arranged in series configuration, or a parallel configuration, or a combination thereof.

(84) Referring to FIG. **9** and FIG. **10**, a stack **141'**, according to an alternative embodiment of the disclosure is shown, the stack **141'** includes an outer frame **142'** and a corrugated structure **143'** arranged inside the outer frame **142'** and supported by the outer frame **142'**. The corrugated structure **143'** defines the plurality of vapor channel **116** having substantially triangular shape. Further, the stack **141'** includes two layers of the membrane **122**, for a first layer **144'** of the membrane **122** arranged on a first side to the corrugated structure **143'** and a second layer **145'** of the membrane **122** arranged on a second side, opposite to the first side, of the corrugated structure **143'**. As shown, the first layer **144'**, the second layer **145'**, and the corrugated structure **143'** are arranged such that the corrugated structure **143'** is sandwiched between the first layer **144'** and the second layer **145'**. Also, the first layer **144'** and the second layer **145'**, respectively, abuts the first side of the corrugated structure **143'** and the second side of the corrugated structure **143'**.

Accordingly, the air channels **114** are separated from the adjacent vapor channels **116** by the membrane **122**. In embodiments, the outer frame **142'** and the corrugated structure **143'**, each is made of aluminum. In such a case, the outer frame **142'**, the corrugated structure **143'**, the layers **144'**, **145'** are fastened together by using a plurality of bolts **147'** and one or more cover strips **148'**. In embodiments, one or more gaskets **149'** may be arranged between the cover strip **148'** and the outer frame **142'** to prevent a leakage of vacuum. Further, each vapor channel **116** includes an

outlet to facilitate a fluid communication of the vapor channels **116** with the conduit **140** (shown in FIG. 4). The stack **141'** may include suitable structures, such as a cavity formed in the outer frame **142'**, to receive the conduit **140** and other suitable structures to facilitate the fluid communication between the vapor channels **116** and the conduit **140**. The plurality of stacks **141'** may be assembled similar to the assembly of the stacks **141** to form a dehumidification core.

(85) Referring to FIG. 11 and FIG. 12, a stack **141''**, according to yet another embodiment is shown. In this embodiment, the stack **141''** includes an outer frame **142''** and a corrugated structure **143''** integrally formed with the outer frame **142''** and may be made of plastic. In such a case, the outer frame **142''** and the corrugated structure **143''** are formed using thermoforming. Although thermoforming is contemplated, it may be appreciated that the stack may be formed by using other suitable techniques, such as, but not limited to, extrusion, injection molding, stamping, etc. As shown the outer frame **142''** includes a plurality of flanges **147''** to facilitate at attachment of number of stacks **141''** together to form a dehumidification core. The number of stacks **141''** are arranged substantially parallel to and spaced apart from each in an arrangement similar to the arrangement of the stacks **141**. The stacks **141''** are engaged with each other by inserting fasteners (not shown) though the holes **148''** defined by the flanges **147''**. Further, the stacks may be assembled/connected to each other using a plurality of fasteners. Moreover, one or more spacers may be inserted between two adjacent stacks to form air channel between the two adjacent stacks.

(86) Further, the stack **141''** includes two layers of the membrane **122**, for a first layer **144''** of the membrane **122** arranged on a first side to the corrugated structure **143''** and a second layer **145''** of the membrane **122** arranged on a second side, opposite to the first side, of the corrugated structure **143''**. As shown, the first layer **144''**, the second layer **145''**, and the corrugated structure **143''** are arranged such that the corrugated structure **143''** is sandwiched between the first layer **144''** and the second layer **145''**. Also, the first layer **144''** and the second layer **145''**, respectively, abuts the first side of the corrugated structure **143''** and the second side of the corrugated structure **143''**. In embodiments, the first layer **144''** and the second layer **145''** may be integrally formed with the outer frame **142''** during thermoforming. Moreover, each vapor channel **116** includes an outlet to facilitate a fluid communication of the vapor channels **116** with the conduit **140** (shown in FIG. 4). The stack **141''** may include suitable structures, such as a cavity, to receive the conduit **140** and other suitable structures to facilitate the fluid communication between the vapor channels **116** and the conduit **140**.

(87) Referring to FIGS. 13, 14, and 15, a stack **141'''**, according to yet another embodiment is shown. The stack **141'''** includes a frame structure **1000** having a mesh **1002** and an outer frame **142'''** connected to the sides of the mesh **1000** for supporting the mesh **1002**. Further, the stack **141'''** includes a first layer **144'''** of the membrane **122** attached to the outer frame **142'''** and arranged parallel to the mesh **1002** and spaced apart from the mesh **1002**, and a second layer **145'''** of the membrane **122** attached to the outer frame **142'''** and arranged spaced apart and parallel to the mesh **1002**. Also, an outer edge of the first layer **144'''** and an outer edge of the second layer **145'''** are fused with the outer frame **142'''** to provide a leak proof joint.

(88) Moreover, the first layer **144'''** and the second membrane **145'''** are disposed on the opposite sides of the mesh **1002** such that the mesh **1002** is arranged between the two layers **144'''**, **145'''**. Accordingly, a gap between the two layers **144'''**, **145'''** of the stack **141'''** define the vapor channel **116** therebetween. Also, the mesh **1002** arranged between the layers **144'''**, **145'''** prevents a collapsing of the layers **144'''**, **145'''** when vacuum or low pressure is created within the vapor channel **116**. In embodiments, an outer edge of the first layer **144'''** and an outer edge of second layer **145'''** are press fused to the outer frame **142'''** to provide a leak proof joint. In embodiments, the outer frame **142'''** is made of a thermoplastic material, a rubber, or any other suitable material to enable a fusing of the outer edges of the first layer **144'''** and the second layer **145'''** with the outer frame **142'''** and an outer edge of the mesh **1002**. Further, the membrane **122** (the first layer or the second layer) defines an outlet **138'''** to facilitate a fluid communication with the conduit **140** and

the vapor channel **116**. The stack **141'''** may include suitable structures, such as a cavity formed in the outer frame **142'''**, to receive the conduit **140** and other suitable structures to facilitate the fluid communication between the vapor channels **116** and the conduit **140** via the outlet **138'''**. The plurality of stacks **141'''** may be assembled similar to the assembly of the stacks **141** to form a dehumidification core.

(89) Referring to FIGS. **16**, **17**, and **18**, a dehumidification core **1600**, according to an alternative embodiment is shown. As shown, the dehumidification core **1600** includes a cuboidal structure having a first side face **1602**, a second side face **1604**, a third side face **1606**, a fourth side face **1608**, a top face **1610**, and a bottom face **1612**. The dehumidification core **1600** includes an outer frame **1616**, a plurality of first corrugated structures **1618** defining a plurality of air channels **1620**, and a plurality of second corrugated structures **1622** defining a plurality of vapor channels **1624**. The first corrugated structures **1618** and the second corrugated structures **1622** are supported by the outer frame **1616**, and are disposed within a space defined by the outer frame **1620**. As shown, the outer frame **1616** is disposed along the corners of the cuboidal structure and covers the top face **1610** and the bottom face **1612**.

(90) Further, the air channels **1620** extend from the first side face **1602** to the second side face **1604**, while the vapor channels **1624** extend from the third side face **1606** to the fourth side face **1608**. Accordingly, the air channels **1618** and the vapor channels **1624** extend in a mutually perpendicular direction. It may be appreciated that the air may enter the air channels **1618** through the first side face **1602** and leaves the dehumidification core **1600** through the second side face **1604**. Also, the vapor channels **1624** are closed at the third side face **1606** and are open on the fourth side face **1608**. Therefore, the moisture (i.e., water vapors) extracted from the air leaves the dehumidification core **1600** through the fourth side face **1608**. As shown in FIG. **17**, the first corrugated structures **1618** and the second corrugated structures **1622** are arranged inside the dehumidification core **1600** such that each first corrugated structure **1618** is arranged adjacent/neighboring the second corrugated structure **1622**. In this manner, a single second corrugated structure **1622** is arranged between two adjacent first corrugated structures **1618**. Further, each first corrugated structure **1618** is separated from the adjacent second corrugated structures **1622** by a membrane **1630** such that the membrane **1630** abuts both the first corrugated structure **1618** and the second corrugated structure **1622**. Accordingly, each air channel **1620** is disposed adjacent to one or more of the vapor channels **1624** and is separated from the adjacent vapor channels **1624** by the membrane **1630** (as shown in FIGS. **17** and **18**). The membrane **1630** is identical to the membrane **122** in compositions and function and facilitates a flow of moisture (i.e., water vapors) from the air flowing through the air channels **1620** to the vapor channels **1624** when a low pressure is created/maintained inside the vapor channels **1624**. Further, a conduit (not shown) similar to the conduit **140** may extend from the dehumidification core **1600** to a water ejector to facilitate a fluid communication between the vapor channels **1624** and the water ejector.

(91) Referring to FIGS. **19** and **20**, a dehumidification core **1900**, according to an alternative embodiment is shown. The dehumidification core **1900** is formed by using a conduit **1902** having a wall made of a membrane **1904**. The conduit **1902** may be arranged in a heat exchanger manner to facilitate a flow of air over the conduit **1902**. Further, a channel **1906** of the conduit **1902** defines a vapor channel **1908** of the dehumidification core **1900**. It may be appreciated that one end of the conduit **1902** is closed, while a vacuum or low pressure is created inside the conduit **1902** through an opening defined at another end to facilitate a flow of moisture from the air to the vapor channel **1908** through the membrane **1904**.

(92) A working of the air-cooling system **100** having the dehumidification core **110** is now explained. It may be envisioned that an air-cooling system having the dehumidification core **1600**, **1900** may also function similarly. The air-cooling system **100** receives the first airstream **200A** having relatively higher humidity level and higher temperature from the room **300** and supplies the second airstream **200B** having relatively less humidity level and less temperature to the room **300**.

For so doing, the air-cooling system **100** receives the first airstream **200A** from the room **300** via the inlet opening **706**. In embodiments, the first fan **504** may facilitate a suction/intake of the first airstream **200A** into the first unit **700** via the inlet opening **706**. Upon entering the first unit **700**, the first airstream **200A** passes through the first evaporation pad **500** (i.e., the first evaporative cooler **104**), and in so doing, the first airstream **200A** may cause evaporation of the liquid water disposed inside the first evaporative pad **500**, causing a conversion of liquid water into water vapors. Accordingly, a temperature of the first airstream **200A** decreases upon passing through the first evaporative pad **500** (i.e., the first evaporative cooler **104**) as a latent heat needed to convert the liquid water into water vapors is provided by the first airstream **200A**, while a humidity level of the first airstream **200A** increases upon passing through the first evaporative pad **500**. Therefore, the first evaporative cooler **104** receives the first airstream **200A** having relatively high temperature and low humidity and expels the inlet air **400A** having relatively lower temperature and higher humidity.

(93) Subsequently, the inlet air **400A** enters the dehumidification core **110**, flows through the air channels **114**, and exits the dehumidification core **110** as the outlet air **400B**. During the flow of the inlet air **400A** through the air channels **114**, at least a portion of the water vapors **120** present into the inlet air **400A** flows through the membrane **122** and inside the adjacent vapor channels **116**. For facilitating the extraction of the water vapors **120** from the inlet air **400A**, and the movement of the water vapors **120** inside the vapor channels **116**, a pressure differential is created. The pressure differential is created by creating or maintaining a relatively lower pressure inside the vapor channels **116** as compared to the pressure inside the air channels **114**. In fact, to ensure a flow of the water vapors **120** across the membrane **122**, the partial pressure of water vapors inside the vapor channels **116** is maintained at a lower value relative to the partial pressure of water vapors inside the air channels **114**. For so doing, the controller may control and operates the pump **152** so as to pump the liquid water from the reservoir **154** to the inlet port **160** of the water ejector **150** at an appropriate speed (i.e., a predetermined quantity of the liquid water enters the inlet portion **158** per second). The appropriate speed of the liquid water entering the water ejector **150** may be determined based on the humidity level of the room **300** and/or the humidity level of the inlet air **400A** and/or a speed and amount of first airstream **200A** entering the air-cooling system **100**.

(94) As the liquid water flows through the inlet portion **158** and enters the throat portion **162**, the speed of the liquid water increases, and reaches a maximum value at the throat portion **162**. Consequently, a relatively low pressure (i.e., vacuum) is created at the throat portion **162**, and hence a relatively low pressure is created inside the vapor channels **116** as the vapor channels **116** are in fluid communication with the throat portion **162** via the vapor inlet port **168** and the conduit **140**. It may be appreciated that a value of pressure inside the vapor channels **116** may be relatively higher as compared to a value of pressure inside the throat portion **162** due to loss of vacuum caused by the length of conduit **140** and any other bends in the conduit **140**. Further, the pressure in the throat portion **162** is created and maintained such that the pressure inside the vapor channels **116** is lower and at a desired pressure differential than the pressure inside the air channels **114**. Owing to the differential pressure between the vapor channels **116** and the air channels **114**, the water vapors **120** are extracted from the air flowing through the air channels and moves across the membrane **122** to the vapor channels **116**. Similarly, the water vapors **120** extracted from the inlet air **400A** may move/flow through the conduit **140** and enters the throat portion **162** via the vapor inlet port **168** due the pressure differential between the vapor channels **116** and the throat portion **162**. Upon entering the water ejector **150** (i.e., the throat portion **162**), the water vapors **120** may move along the liquid water and enters the outlet portion **164** of the water ejector **150**. Due to the increase of the cross-sectional area in the outlet portion **164**, the velocity of the liquid water decreases, thereby creating a relative high pressure inside the outlet portion **164** as compared to the pressure inside the throat portion **162**. Accordingly, the water vapors **120** received from the vapor channels **116** condense inside the outlet portion **164** before leaving the water ejector **150**. Accordingly, a

need for a separate condenser for condensing the water vapors **120** into the liquid water is eliminated, thereby increasing an efficiency of the dehumidification system **102**, and hence the air-cooling system **100**.

(95) Heat may be generated due to the condensation of the water vapors **120** inside the outlet portion **164** of the water ejector **150**, resulting into an increase in temperature of the liquid water exiting the outlet port **166** of the water ejector **150**. To reduce the temperature of the liquid water before delivering the liquid water to the reservoir **154**, at least a portion of the liquid water is directed to the heat exchanger **182** that facilitates the cooling of the received liquid water. Thereafter, the cooled liquid water is supplied to the reservoir **154** for storage and is available for subsequent supply to the water ejector **150** by the pump **152**. It may be noted that a volume of the liquid water exiting the water ejector **150** and received by the reservoir **154** may be higher than a volume of the liquid water supplied by the pump **152** to the water ejector **150** due to the condensation of the water vapors **120** received from the dehumidifier core **110**. Accordingly, the level of the liquid water inside the reservoir **154** may increase. For preventing an overflow of the reservoir **154** and spilling of the liquid water from the reservoir **154**, the drain valve **180** may be opened when the level of the liquid water inside the reservoir **154** is above a first threshold level. Opening of the drain valve **180** may allow a drainage of the liquid water from the reservoir **154**. Further, the drain valve **180** may be closed in response to a decrease of the level of the liquid water below a second threshold value. In certain implementations, the drain valve **180** is adapted to automatically opened and closed, respectively, in response to the level of the liquid water reaching above the first threshold value and falling below the second threshold value.

(96) Further, the outlet air **400B** expelled from the dehumidifier core **110** is received by the second evaporative cooler **106**, gets cooled, and exit the second evaporative cooler **106** as the second air stream **200B**. Upon entering the second evaporative cooler **106**, the outlet air **400B** passes through the second evaporation pad **600**, and in so doing, the outlet air **400B** may cause the evaporation of liquid water disposed inside the second evaporative pad **600**, causing a conversion of liquid water into water vapors. Accordingly, a temperature of the air decreases upon passing through the second evaporative pad **600** as a latent heat needed to convert the liquid water into water vapors is provided by the outlet air **400B**, while a humidity level of the air increases upon passing through the second evaporative pad **600**. Therefore, the second airstream **200B** discharged from the second evaporative cooler **106** is relatively cooler than the outlet air **400B** received from the dehumidifier core **110** and has a relatively higher humidity compared to the outlet air **400B**. The second airstream **200B** after exiting the second evaporative cooler **106**, exits the air-cooling system **100** via the outlet opening **708**, and enters the room **300**. It may be appreciated that air-cooling system **100**, in general, and the dehumidifier system **102**, in particular, is controller in a manner such that the second airstream **200B** has relatively less humidity level (i.e., moisture content) relative to the first air stream **200A**. For so doing, the pressure at the throat portion **162**, and hence inside the vapor channels **116**, is controlled such that total amount of moisture extracted by dehumidifier core **110** is greater than a total amount of moisture absorbed by the first airstream **200A** upon passing through the first evaporative cooler **104** and the second evaporative cooler **106**. Further, a utilization of the liquid water as motive fluid for creating low pressure inside the throat portion **162**, and hence inside the vapor channels **116**, facilitates in decreasing energy consumption due to non-compressible nature of the liquid water. Further, a use of the water ejector **150** as a pressure reducing means prevents cavitation related damages relative to a scenario in which a conventional pump is used for creating low pressure or vacuum in the vapor channels **116**. Also, use of the water ejector **116** facilitates in decreasing an overall size of the dehumidification system **102**. Although the dehumidification system **102** is contemplated and explained in conjunction with the air-cooling system **100**, it may be envisioned that the dehumidification system **102** may be utilized as a stand-alone system for removing moisture from an air. Also, an application of the dehumidification system **102** as a water extraction and collection may also be contemplated.

(97) Also, the use of water ejector **150** provides a capability of handling high condensable fraction in the form of water vapor, as well as creating a source of fresh water as it condenses moisture extracted at the dehumidification core **110**. Further the water ejector **150** is capable of generating vacuum pressures as low as 5 mbar absolute (expressed as mbarA. Also, the water ejector can be tailored to operate in a range of vacuum pressures such as 100 mbarA, 150 mbarA or even 500 mbarA or greater. Vacuum pressure is regulated by adjusting fluid variables such as pressure and flow rate.

(98) The patentable scope is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims. All citations referred herein are expressly incorporated herein by reference.

(99) As used herein, the term “include” and its grammatical variants are intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that can be substituted or added to the listed items. The terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Unless otherwise defined, all terms have the same meaning as commonly understood by one of ordinary skill in the art. The singular forms “a,” “an,” and “the,” include plural references unless expressly and unequivocally limited to one referent.

Claims

1. A dehumidification system for removing water vapors from an air, the dehumidification system comprising: at least one dehumidification core defining an air channel and at least one vapor channel separated from the air channel and including a membrane separating the at least one vapor channel from the air channel and adapted to facilitate a removal of moisture from the air flowing through the air channel, wherein the membrane is selectively permeable to water and water vapor and impermeable to air, wherein the membrane includes an ion exchange capacity of at least 1.0 milliequivalents per gram; a liquid ejector having: a throat portion fluidly coupled to the at least one vapor channel and adapted to create a relatively lower pressure of water vapor within the at least one vapor channel than in the air channel, and an outlet portion disposed downstream of the throat portion and configured to increase the pressure inside the liquid ejector to facilitate a condensation of the water vapors received from the at least one vapor channel; a reservoir for storing a liquid and configured to receive the liquid from the liquid ejector; and a pump fluidly connected to the liquid ejector and the reservoir and configured to supply the liquid from the reservoir to the liquid ejector.
2. The dehumidification system of claim 1, wherein the reservoir includes a drain valve for facilitating a drainage the liquid from the reservoir when a level of the liquid inside the reservoir reaches above a first threshold level.
3. The dehumidification system of claim 1, further including a heat exchanger fluidly connected to the liquid ejector and configured to receive at least a portion of the liquid exiting the liquid ejector, wherein the heat exchanger is configured to cool the received liquid.
4. The dehumidification system of claim 3, wherein the heat exchanger is disposed upstream of the reservoir and supply the cooled liquid to the reservoir.
5. The dehumidification system of claim 1, wherein the relatively lower pressure of water vapor is up to -960 mbar gage pressure.
6. The dehumidification system of claim 1, wherein the liquid ejector is a water ejector adapted to receive liquid water from the pump.

7. The dehumidification system of claim 1, wherein the membrane comprises a sulfonated block copolymer having one or more copolymer block configurations corresponding to any of A-B-A, A-B-A-B-A, (A-B-A).sub.nX, (A-B).sub.nX, A-D-B-D-A, A-B-D-B-A, (A-D-B).sub.nX, (A-B-D).sub.nX or mixtures thereof, where n is an integer from 2 to 30, X is a coupling agent residue, and wherein the sulfonated polymer has a degree of sulfonation of >25.
 8. The dehumidification system of claim 1, further including a booster pump arranged between the dehumidification core and the liquid ejector to reduce the pressure inside the vapor channels to a pressure below a partial vapor pressure of water.
 9. The dehumidification system of claim 8, the booster pump facilitates in reducing the pressure to a value less than equal to 31.7 mbarA at 25° C.
 10. The dehumidification system of claim 9, wherein the booster pump facilitates in reducing the pressure to <150 mbarA, or to 10 mbarA-40 mbarA.
 11. An air-cooling system, comprising the dehumidification system of claim 1, and an evaporative cooler for cooling the air by facilitating an absorption of water into the air, wherein the evaporative cooler is disposed downstream or upstream of the at least one dehumidification core.
 12. An air-cooling system, comprising: at least one dehumidification core defining an air channel and at least one vapor channel separated from the air channel and including a membrane separating the at least one vapor channel from the air channel and adapted to facilitate a removal of moisture from the air flowing through the air channel, wherein the membrane is selectively permeable to water and water vapor and impermeable to air, wherein the membrane includes an ion exchange capacity of at least 1.0 milliequivalents per gram; a water ejector having a throat portion fluidly coupled to the at least one vapor channel and adapted to create a relatively lower pressure within the at least one vapor channel than in the air channel, and an outlet portion disposed downstream of the throat portion and configured to increase the pressure of water to facilitate a condensation of the water vapors received from the at least one vapor channel; a reservoir for storing water and configured to receive the water from the water ejector; a pump fluidly connected to the water ejector and the reservoir and configured to supply the water from the reservoir to the water ejector; and an evaporative cooler for cooling the air by facilitating an absorption of water into the air, the evaporative cooler is disposed downstream or upstream of the at least one dehumidification core.
 13. The air-cooling system of claim 12, wherein the reservoir includes a drain valve for facilitating a drainage of the liquid water from the reservoir when a level of the liquid water reaches above a first threshold level.
 14. The air-cooling system of claim 12, further including a heat exchanger fluidly connected to the water ejector and configured to receive at least a portion of the liquid water exiting the water ejector, wherein the heat exchanger is configured to cool the received liquid water.
 15. The air-cooling system of claim 12, wherein the heat exchanger is disposed upstream of a reservoir and supply the cooled liquid water to the reservoir, and wherein the heat exchanger is an air to liquid heat exchanger.
 16. The air-cooling system of claim 12, further including a booster pump arranged between the dehumidification core and the water ejector to reduce the pressure inside the vapor channels to a pressure below a partial vapor pressure of water.
 17. The air-cooling system of claim 16, wherein the booster pump facilitates in reducing the pressure to a value $\leq$ 31.7 mbarA, or <150 mbarA, or between 20 mbarA and 40 mbarA at 25° C.
 18. The air-cooling system of claim 12, wherein the evaporative cooler is a first evaporative cooler disposed upstream of the dehumidification core and adapted to cool the air flowing to the dehumidification core, the air-cooling system further includes a second evaporative cooler disposed downstream of the dehumidification core and configured to cool the air received from the dehumidification core.
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